Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

Academic Year: 2026–27

Subject: Software Engineering (SE) Practical / Oral Examination

Q 1: Draw the Use Case Diagram for a Library Management System.

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Q 2 : Draw the Class Diagram for a Hospital Management System.

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Q 3: Draw the Statechart Diagram for a Online shopping cart

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Q 4: Draw the Activity Diagram for a student Management System.

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Q 5: Estimate the development effort and time for a project of size 20 KLOC using the Basic COCOMO model. Assume the project is Organic, with (a=2.4, b=1.05, c=2.5, d=0.38).Calculate the estimated project cost if the development cost is ₹30,000 per person-month.

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Q 6: A college plans to develop a **Student Management System** to manage student registration, attendance, examination results, and fee details.
1.Identify the major project activities and assign suitable durations.
2.Define task dependencies and prepare a project schedule using a Gantt chart

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Q 7. Consider a Hostel Management System that manages student hostel admissions, room allocation, fee payments, and complaints.
Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

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Q.8.A college plans to develop an Online Examination System to conduct online tests, evaluate answers, generate results, and maintain student records. The requirements may change based on feedback from faculty and students. Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

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Q 9. 1. Apply Basis Path Testing to the given program and calculate the Cyclomatic Complexity.

```
read n
if (n > 0)
   if (n % 2 == 0)
      print "Even"
   else
      print "Odd"
else
   print "Invalid"
```

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Q 10: Apply White-box testing and calculate cyclomatic complexity for given program

```
read n
i = 2; isPrime = true
while (i <= n/2)
if (n % i == 0)
isPrime = false
break
i = i + 1;print isPrime
```

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Q 11: Given the following pseudo-code, apply Basis Path Testing to determine:

```
read a, b
if (a > b)
 max = a
else
 max = b
 if (max > 0)
 print "Positive"
else
print "Non-positive"
```

1. Control Flow Graph (CFG)
2. Cyclomatic Complexity
3. Independent Paths
4. Test Cases

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Q 12: Draw the Component Diagram for a Library Management System.

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Q 13.Hospital Management System

A hospital plans to develop a Hospital Management System to manage patient registration, appointments, billing, laboratory reports, pharmacy, and electronic health records. The hospital requires a secure and reliable system with clearly defined requirements. Major requirement changes are not expected during development.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

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Q. 14 Mobile Banking Application

A bank plans to develop a Mobile Banking Application with fund transfer, balance enquiry, bill payment, investment services, and biometric authentication. Since financial transactions are involved, the system requires high security, extensive testing, and risk management.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

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Q. 15 AI-Based Smart Traffic Management System

A city corporation plans to develop an AI-based Smart Traffic Management System to monitor traffic, optimize signal timings, detect accidents, and provide real-time route suggestions. The project involves AI integration, IoT devices, and evolving requirements.

Study the given case study and identify a suitable software process model. Draw the diagram of the selected process model.

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Q. 16. Consider course registration system

Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

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Q. 17. Consider railway reservation system

Draw the Context-Level DFD (Level 0) for the given case study. Draw the Level 1 DFD, showing the major processes, data stores, external entities, and data flows. Label all processes, data stores, external entities, and data flows appropriately.

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Q. 18. A healthcare startup plans to develop an Online Pharmacy Management System to enable customers to search medicines, upload prescriptions, place orders, make online payments, and track deliveries. The system will also allow pharmacists to verify prescriptions, manage inventory, and generate sales reports.

The estimated software size is 32 KLOC. Based on previous projects, the organization classifies it as a Semi-Detached project.

The function counts identified are:

- External Inputs (EI): 18
- External Outputs (EO): 14
- External Inquiries (EQ): 10
- Internal Logical Files (ILF): 8
- External Interface Files (EIF): 5

Assume all functions are of average complexity and the Value Adjustment Factor (VAF) = 1.05.

Questions:

1. Calculate the Unadjusted Function Points (UFP).
2. Calculate the Adjusted Function Points (AFP).
3. Calculate the project effort using the Basic COCOMO model. a=2.4, b=1.05, c=2.5, d=0.38
4. Find the estimated effort and discuss whether the project is suitable for the selected COCOMO category.

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Q. 19. A municipal corporation plans to develop a Smart Parking Management System that allows users to locate available parking spaces, reserve slots, make online payments, and receive real-time parking updates. Parking administrators can monitor occupancy and generate daily reports.

The estimated project size is 28 KLOC, and the project is categorized as Organic. a=2.4, b=1.05, c=2.5, d=0.38

Questions:

1. Calculate the project effort and development time using the Basic COCOMO model.

2. If the organization's productivity is 600 LOC per person-month, Calculate the effort using the LOC method.

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Q. 20. Draw Class diagram for Travel Agency system